

Force Systems

- ☐ FORCE SYSTEMS
- ☐ MOMENTS OR TORQUE
- ☐ COUPLE MOMENT



NU - FAST

Dr. Kashif Saeed (kashif.saeed@nu.edu.pk)

ME101 (Spring 2017)

Force Systems

Basic Mech. Engg.

Force Systems

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Moments or Torque

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Couple Moment

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Force Systems

VIDEO



[1]

[1] - Youtube accessed on February 22, 2015 (<https://www.youtube.com/watch?v=MfNz7DRwtQQ>)

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1 / 29

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NEWTON'S LAWS OF MOTION

What is Newton's First Law of Motion?

A body at rest remains at rest and a body in motion continues its motion in a straight line unless acted upon by an external force

What is Newton's Second Law of Motion?

The force action on a body gives it an acceleration which is in the direction of force

$$F = m a$$

What is Newton's Third Law of Motion?

If one object exerts a force on another object, the other exerts the same force in the opposite direction on the one

Action = - Reaction
Contact force

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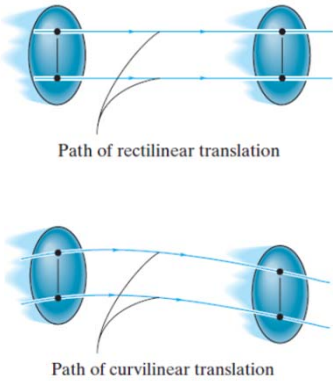
Couple Moment
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Force Systems

FORCE & MOTION

What kind of motion can a force generate in a rigid body?

Translation



Path of rectilinear translation

Path of curvilinear translation [1]

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 312

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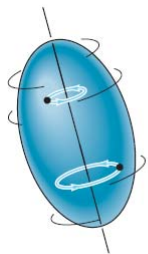
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FORCE & MOTION

What kind of motion can a force generate in a rigid body?

Translation

Rotation about a fixed axis



Rotation about a fixed axis [1]

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 312

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FORCE & MOTION

What kind of motion can a force generate in a rigid body?

Translation

Rotation about a fixed axis

General Motion



General plane motion [1]

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 312

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●●●●●●●●●●

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MOMENTS OR TORQUE

What is the tendency of a force to produce rotary motion in a body called?

Moment or Torque

What are the factors on which the magnitude of moment depends?

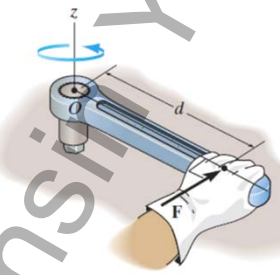
Magnitude of force, F

Perpendicular distance from axis of rotation d

$$M = F d$$

The magnitude of the moment is directly proportional to the magnitude of Force F and the perpendicular distance or moment arm d

Moment is defined with respect to a fix reference point



[1]

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 117

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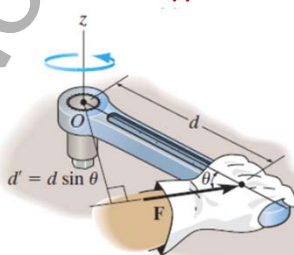
Moments or Torque
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Couple Moment
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Moments or Torque

BASIC CONCEPTS

What if the force is applied at an angle?



$d' = d \sin \theta$

The component of distance is taken, which is perpendicular to the force

Or the component of force is taken, which is perpendicular to the distance

[1]

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 117

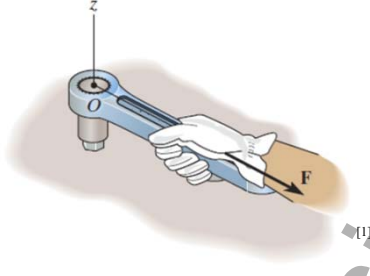
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Moments or Torque

BASIC CONCEPTS

What if the force is passing through the reference point?



No Moment

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 117

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VECTOR NOTATION

Is moment a scalar or a vector?

Vector

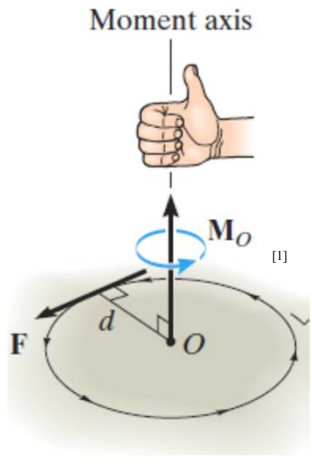
What about its direction?

It is perpendicular to the plane that contains the force and its moment arm

Which mathematical operator, when applied to two vectors, give an other vector which is perpendicular to both?

$$\underline{M_o} = \underline{r} \times \underline{F}$$

r can be any position vector, measured from point O to any point on the line of action of the force F



[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 118

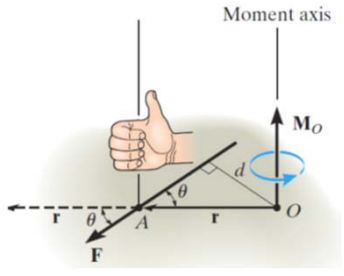
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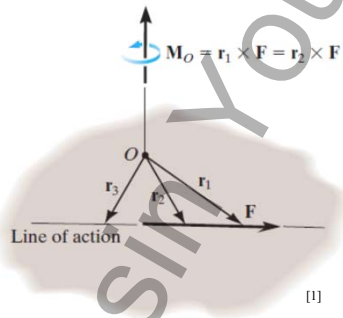
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Couple Moment
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Moments or Torque

PRINCIPLE OF TRANSMISSIBILITY

$$\underline{M}_O = \underline{r} \times \underline{F}$$

F can be applied at any point along its line of action and still create this same moment about point O

This property is called the principle of transmissibility of a force

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp. 118

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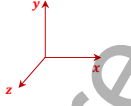
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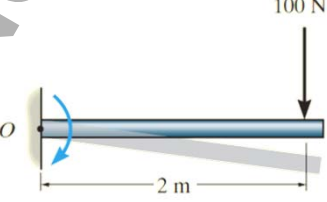
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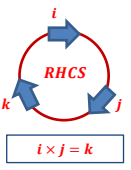
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Moments or Torque

EXAMPLE I

Calculate moment about point O

Magnitude:

$$M_o = F d = (100)(2) = 200 \text{ Nm}$$

Direction:

-ve z-axis

$$\underline{r} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \text{ m} \quad \underline{F} = \begin{bmatrix} 0 \\ -100 \\ 0 \end{bmatrix} \text{ N}$$
$$\underline{M}_o = \underline{r} \times \underline{F}$$
$$\underline{M}_o = \begin{bmatrix} 0 \\ 0 \\ -200 \end{bmatrix} \text{ Nm}$$

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11 / 29

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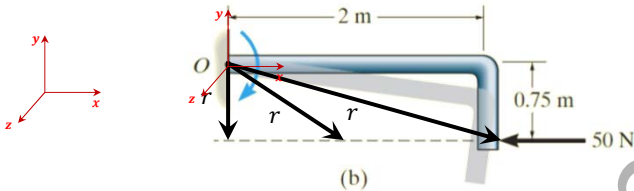
Couple Moment
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Moments or Torque

EXAMPLE II

Calculate moment about point O



(b)

Magnitude:

$$M_o = F d = (50)(0.75) = 37.5 \text{ Nm}$$

Direction:

-ve z-axis

$$\underline{r} = \begin{bmatrix} 2 \\ -0.75 \\ 0 \end{bmatrix} \text{ m} \quad \underline{F} = \begin{bmatrix} -50 \\ 0 \\ 0 \end{bmatrix} \text{ N}$$
$$\underline{M}_o = \underline{r} \times \underline{F}$$
$$\underline{M}_o = \begin{bmatrix} 0 \\ 0 \\ -37.5 \end{bmatrix} \text{ Nm}$$

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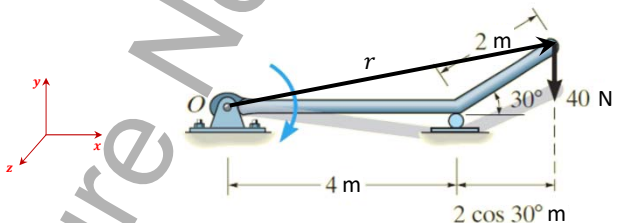
Couple Moment
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Moments or Torque

EXAMPLE III

Calculate moment about point O



4 m

2 m

30°

40 N

2 cos 30° m

Magnitude:

$$M_o = F d = (40)(4 + 2 \cos(30^\circ)) = 229.282 \text{ Nm}$$

Direction:

-ve z-axis

$$\underline{r} = \begin{bmatrix} 5.732 \\ 1 \\ 0 \end{bmatrix} \text{ m} \quad \underline{F} = \begin{bmatrix} 0 \\ -40 \\ 0 \end{bmatrix} \text{ N}$$
$$\underline{M}_o = \underline{r} \times \underline{F}$$
$$\underline{M}_o = \begin{bmatrix} 0 \\ 0 \\ -229.283 \end{bmatrix} \text{ Nm}$$

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13 / 29

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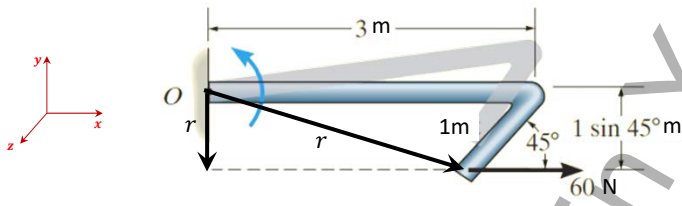
Couple Moment
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Moments or Torque

EXAMPLE IV

Calculate moment about point O



Magnitude:

$$M_o = F d = (60)(1 \sin(45^\circ)) = 42.426 \text{ Nm}$$

Direction:

+ve z-axis

$$\underline{r} = \begin{bmatrix} 3 - \cos 45^\circ \\ -\sin 45^\circ \\ 0 \end{bmatrix} \text{ m} \quad \underline{F} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix} \text{ N}$$
$$\underline{M}_o = \underline{r} \times \underline{F}$$
$$\underline{M}_o = \begin{bmatrix} 0 \\ 0 \\ 42.426 \end{bmatrix} \text{ Nm}$$

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14 / 29

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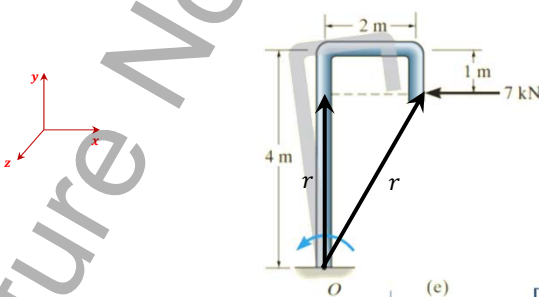
Couple Moment
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Moments or Torque

EXAMPLE V

Calculate moment about point O



Magnitude:

$$M_o = F d = (7000)(4 - 1) = 21 \text{ kNm}$$

Direction:

+ z-axis

$$\underline{r} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \text{ m} \quad \underline{F} = \begin{bmatrix} -7000 \\ 0 \\ 0 \end{bmatrix} \text{ N}$$
$$\underline{M}_o = \underline{r} \times \underline{F}$$
$$\underline{M}_o = \begin{bmatrix} 0 \\ 0 \\ 21000 \end{bmatrix} \text{ Nm}$$

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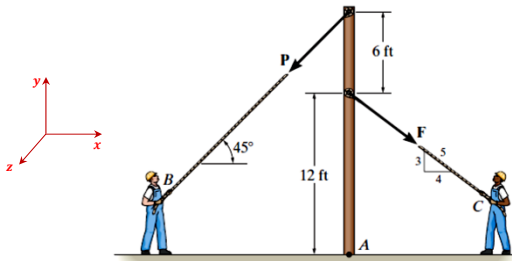
Couple Moment
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Moments or Torque

QUIZ ON A1

Calculate moment about point A

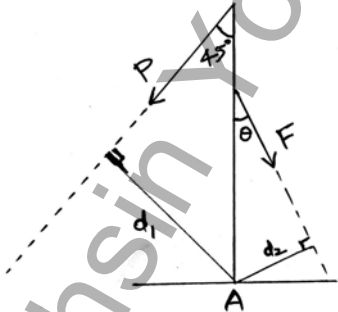


Magnitude:

$$M_A = P d_1 - F d_2$$
$$d_1 = 18 \sin 45^\circ$$
$$d_2 = 12 \left(\frac{4}{5}\right)$$
$$M_A = 636.39 - 768 = -131.6 \text{ Nm}$$

Direction:

- z-axis (clockwise)



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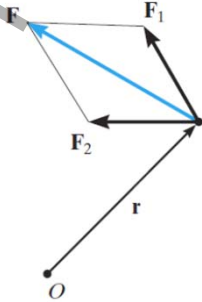
Couple Moment
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Moments or Torque

PRINCIPLE OF MOMENTS OR VARIGNON'S THEOREM

the moment of a force about a point is equal to the sum of the moments of the components of the force about the point





Pierre Varignon
(1654-1722)

$$\underline{M}_0 = \underline{r} \times \underline{F} = \underline{r} \times (\underline{F}_1 + \underline{F}_2)$$
$$\underline{M}_0 = \underline{r} \times \underline{F} = \underline{r} \times \underline{F}_1 + \underline{r} \times \underline{F}_2$$

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16 / 29

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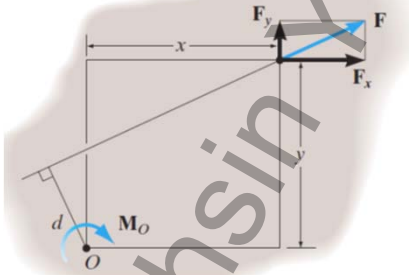
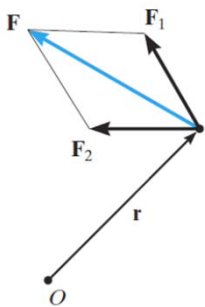
Couple Moment
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Moments or Torque

PRINCIPLE OF MOMENTS OR VARIGNON'S THEOREM

the moment of a force about a point is equal to the sum of the moments of the components of the force about the point



$$M_O = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2) = \mathbf{r} \times \mathbf{F}_1 + \mathbf{r} \times \mathbf{F}_2$$

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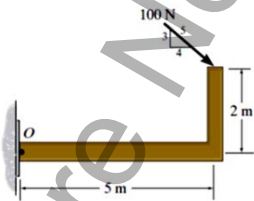
Couple Moment
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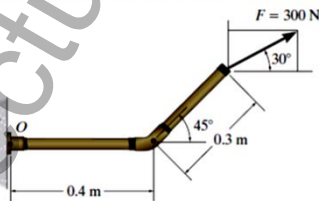
Moments or Torque

PRINCIPLE OF MOMENTS OR VARIGNON'S THEOREM (FUNDAMENTAL PROB)

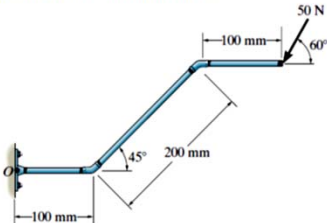
F4-2. Determine the moment of the force about point *O*.



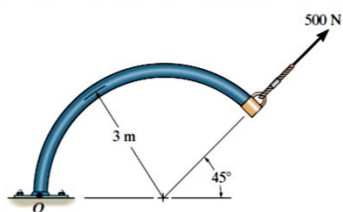
F4-3. Determine the moment of the force about point *O*.



F4-5. Determine the moment of the force about point *O*. Neglect the thickness of the member.



F4-6. Determine the moment of the force about point *O*.



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Moments or Torque
●●●●●●●●●●

Couple Moment
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Moments or Torque

PRINCIPLE OF MOMENTS OR VARIGNON'S THEOREM (FUNDAMENTAL PROB)

F4-7. Determine the resultant moment produced by the forces about point *O*.

F4-8. Determine the resultant moment produced by the forces about point *O*.

F4-10. Determine the moment of force *F* about point *O*. Express the result as a Cartesian vector.

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Moments or Torque

EXAMPLE VI

Calculate the resultant moment about point *O*

Formulation:

$$F_1 = 50\text{ N}, F_2 = 60\text{ N}, F_3 = 20\text{ N}, F_4 = 40\text{ N}$$
$$d_1 = 2\text{ m}, d_2 = 0\text{ m}, d_3 = 3 \sin 30^\circ \text{ m},$$
$$d_4 = 4 + 3 \cos 30^\circ \text{ m}$$

Magnitude:

$$M_o = F_1 d_1 + F_2 d_2 + F_3 d_3 + F_4 d_4$$
$$= -(50)(2) + 60(0) + 20(3 \sin 30^\circ)$$
$$- (40)(4 + 3 \cos 30^\circ) = -333.923 \text{ Nm}$$

Direction:

-ve z-axis

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19 / 29

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Force Systems

Moments or Torque

Couple Moment

Moments or Torque

EXAMPLE VI

Calculate the resultant moment about point O

Formulation:

$$\underline{r}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \text{ m} \quad \underline{F}_1 = \begin{bmatrix} 0 \\ -50 \\ 0 \end{bmatrix} \text{ N}$$

$$\underline{r}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ m} \quad \underline{F}_2 = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix} \text{ N}$$

$$\underline{r}_3 = \begin{bmatrix} 4 + 3 \cos 30^\circ \\ -3 \sin 30^\circ \\ 0 \end{bmatrix} \text{ m} \quad \underline{F}_3 = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix} \text{ N}$$

$$\underline{r}_4 = \begin{bmatrix} 4 + 3 \cos 30^\circ \\ -3 \sin 30^\circ \\ 0 \end{bmatrix} \text{ m} \quad \underline{F}_4 = \begin{bmatrix} 0 \\ -40 \\ 0 \end{bmatrix} \text{ N}$$

Solution:

$$\underline{M}_r = \underline{M}_1 + \underline{M}_2 + \underline{M}_3 + \underline{M}_4$$

$$\underline{M}_r = \underline{r}_1 \times \underline{F}_1 + \underline{r}_2 \times \underline{F}_2 + \underline{r}_3 \times \underline{F}_3 + \underline{r}_4 \times \underline{F}_4$$

$$\underline{M}_r = \begin{bmatrix} 0 \\ 0 \\ -333.923 \end{bmatrix} \text{ Nm}$$

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Moments or Torque

Couple Moment

Moments or Torque

ABOUT A SPECIFIED AXIS

Determine the resultant moment of the three forces about x-axis, the y-axis, and the z-axis.

$M_x = (60\text{N})(2\text{m}) + (50\text{N})(2\text{m}) + 0 = 220\text{Nm}$

$M_y = 0 - (50\text{N})(3\text{m}) - (40\text{N})(2\text{m}) = -230\text{Nm}$

$M_z = 0 + 0 - (40\text{N})(2\text{m}) = -80\text{Nm}$

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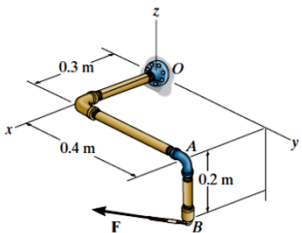
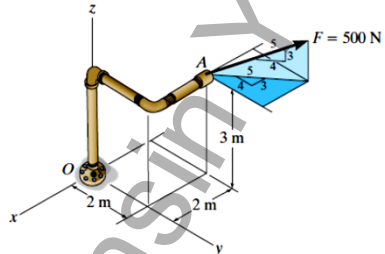
Moments or Torque

ABOUT A SPECIFIED AXIS (FUNDAMENTAL PROB)

F4-13. Determine the magnitude of the moment of the force $\mathbf{F} = \{300\mathbf{i} - 200\mathbf{j} + 150\mathbf{k}\}$ N about the x axis. Express the result as a Cartesian vector.

F4-14. Determine the magnitude of the moment of the force $\mathbf{F} = \{300\mathbf{i} - 200\mathbf{j} + 150\mathbf{k}\}$ N about the OA axis. Express the result as a Cartesian vector.

F4-18. Determine the moment of force $\mathbf{F}$ about the x , the y , and the z axes. Use a scalar analysis.

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Couple Moment

COUPLE MOMENT

Calculate the resultant moment

$$M_o = (30)(0.2) + (30)(0.2)$$

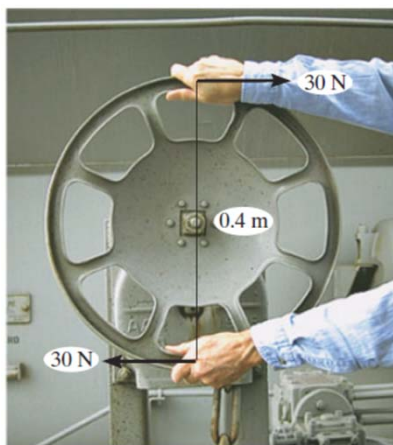
$$M_o = (30)(0.2 + 0.2)$$

$$M_o = (30)(0.4)$$

$$M_o = 12 \text{ Nm}$$

What generates a couple moment?

Two parallel forces that have the same magnitude, but opposite directions, and are separated by a perpendicular distance

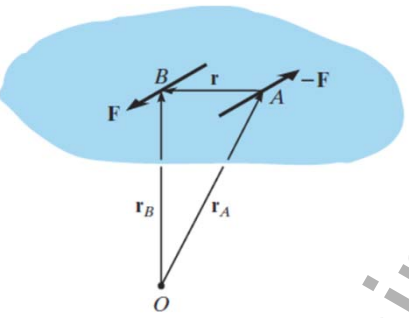


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Couple Moment

COUPLE MOMENT



$$\mathbf{M} = \mathbf{r}_B \times \mathbf{F} + \mathbf{r}_A \times -\mathbf{F} = (\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{F}$$

$$\mathbf{r} = \mathbf{r}_B - \mathbf{r}_A$$

$$\mathbf{M} = \mathbf{r} \times \mathbf{F}$$

A couple moment is a free vector
It can act at any point and would result the same

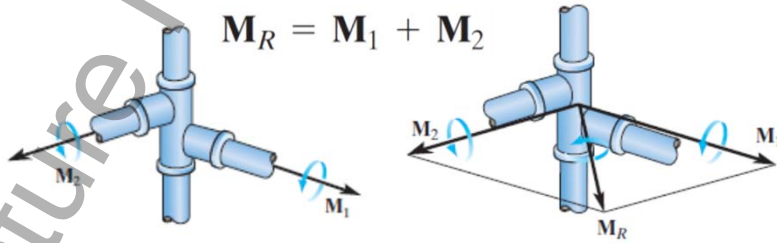
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RESULTANT

resultant couple moment is simply the vector sum of all the couple moments of the system.



$$\mathbf{M}_R = \mathbf{M}_1 + \mathbf{M}_2$$

For more than two vectors $\mathbf{M}_R = \Sigma(\mathbf{r} \times \mathbf{F})$

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Basic Mech. Engg.

Couple Moment

IMPORTANT POINTS

- A couple moment is produced by **two non-collinear forces** that are **equal in magnitude** but **opposite in direction**.
- A couple moment is a **free vector**, and as a result it causes the same rotational effect on a body regardless of where the couple moment is applied to the body.
- The moment of the two couple forces can be determined about **any point**. For convenience, this point is often chosen on the **line of action of one of the forces** in order to eliminate the moment of this force about the point.
- In **three dimensions** the couple moment is often determined using the vector formulation, $\mathbf{M} = \mathbf{r} \times \mathbf{F}$, where $\mathbf{r}$ is directed from **any point** on the line of action of one of the forces to **any point** on the line of action of the other force $\mathbf{F}$.
- A resultant couple moment is simply the **vector sum of all the couple moments** of the system.

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Moments or Torque

Couple Moment

Couple Moment

EXAMPLE VIII

Determine the couple moment acting on the pipe

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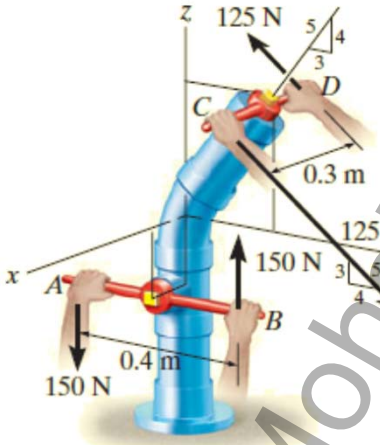
Moments or Torque
●●●●●●●●

Couple Moment
●●●●

Couple Moment

EXAMPLE IX

Determine the moment acting on the pipe



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ME101 (Spring 2017)

28 / 29

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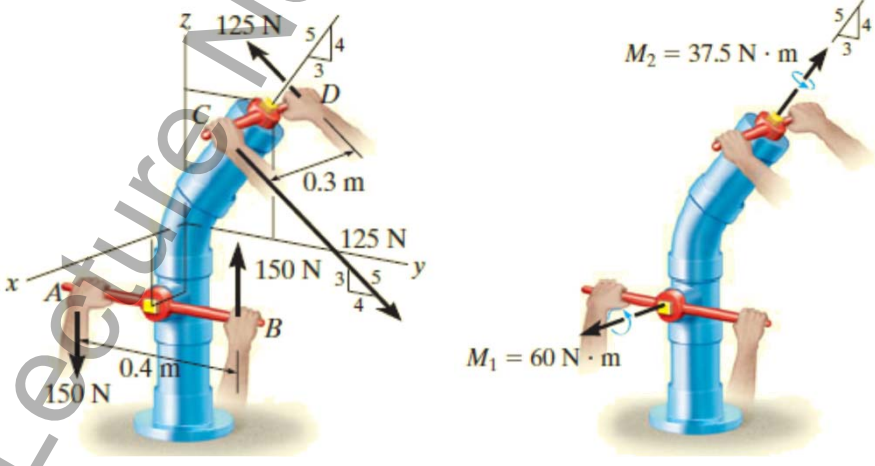
Moments or Torque
●●●●●●●●

Couple Moment
●●●●

Couple Moment

EXAMPLE IX

Determine the moment acting on the pipe

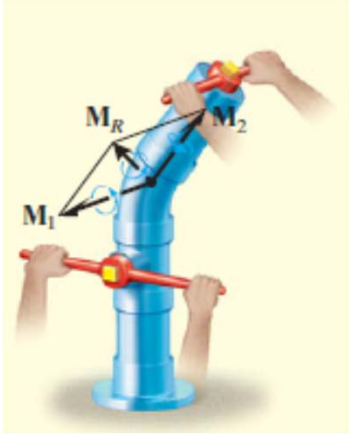


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ME101 (Spring 2017)

29 / 29


$$M_R = M_1 + M_2 = \begin{bmatrix} 60 \\ 22.5 \\ 30 \end{bmatrix} \text{ Nm}$$

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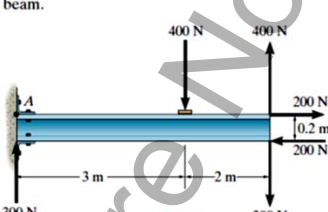
Moments or Torque
●●●●●●●●

Couple Moment
●●●●

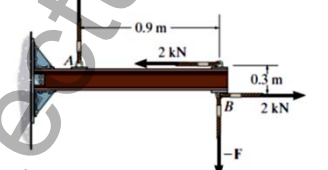
Couple Moment

FUNDAMENTAL PROBLEMS

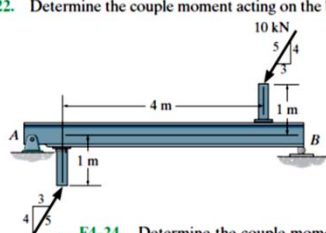
F4-19. Determine the resultant couple moment acting on the beam.



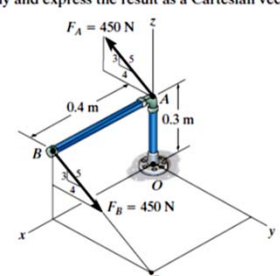
F4-21. Determine the magnitude of F so that the resultant couple moment acting on the beam is 1.5 kN · m clockwise.



F4-22. Determine the couple moment acting on the beam.



F4-24. Determine the couple moment acting on the pipe assembly and express the result as a Cartesian vector.



Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp 154

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17 / 29